

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

2. Subject Name: Intermediates and Transition States in Organic Synthesis

L-T-P: 3-0-0

Credits:3

3. Pre-requisites: Nil

4. Syllabus and reference books:

Module 1: (8L)

Other reactive intermediates carbene, nitrene and free radicals: Electronic Structures, generation, reactivity and reactions.

Module 2: (6L)

Molecular Rearrangements (skeletal rearrangement towards electron deficient C) and their strategic applications: Wagner-Meerwein rearrangement, Favorskii rearrangement, semipinacolone rearrangement, Benzil-benzilic acid rearrangement, Dienone phenol, Demjanov rearrangement, Tiffeneau–Demjanov rearrangement, Cascade rearrangements.

Module 3: (8L)

Chemistry of phosphorus, sulphur, silicon in organic chemistry: Various phosphorus ylides and their detail reactions with mechanism, Various sulfonium ylides and their detail reactions with mechanism, comparison of silicon and carbon, stability of positive charge at β -carbon, *ipso*-substitution of aryl silane, silanes as nucleophiles.

Module 4: (10L)

Polycyclic aromatic hydrocarbons: Basic chemistry of polycyclic aromatic hydrocarbons, synthesis and reactions of naphthalene, anthracene, phenanthrene. Heterocyclic compounds having one heteroatom: Furan, pyrrole, thiophene, pyridine, quinoline, isoquinoline; Reactivity, synthesis and their important reactions with mechanism.

Module 5: (8L)

Pericyclic reactions: Electrocyclic reactions ($4n$ and $4n+2$ electrons systems), Woodward-Hoffman rules, FMO mechanism, Reactions, Cycloaddition: Diels-Alder reactions- FMO mechanism for endo- and exo-selectivity, stereochemistry, regioselectivity, inter and intramolecular reactions, [2+2]-cycloadditions and 1,3-dipolar cycloadditions; Sigmatropic rearrangement: FMO mechanism for reactions [(1,n), (3,3), (2,3) type].

Reference Books:

1. Organic Chemistry 2nd Edition, by J. Clayden, S. Warren, N. Greeves, P. Wothers, Oxford University Press, ISBN-13 978-0199270293.
2. Advanced Organic Chemistry (Part B: Reactions and synthesis) by F. A. Carey and R. J. Sundberg, 5th Corrected ed, 2008, ISBN-13 : 978-0387683546.
3. Modern Methods of Organic Synthesis by William Carruthers and Ian Coldham, Cambridge

University Press; 4th edition, 2004, ISBN-13 : 978-0521778305.

4. Heterocyclic Chemistry, 5th Edition, 2010, by John A. Joule, Keith Mills, ISBN: 978-1-405-13300-5.

5. Strategic Applications of Named Reactions in Organic Synthesis, by Barbara Czako and Laszlo Kurti, Academic Press; 1st edition 2005, ISBN-13: 978-0124297852.

6. Organic Chemistry Vol 1 by I. L. Finar, Pearson Education India; 6th edition, 2002, ISBN-13 : 978-8177585421.

7. March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, Wiley–Blackwell; 6th edition, ISBN-13 : 978-0471720911.

8. Molecular Orbitals and Organic Chemical Reactions, Wiley; 1st edition 2011, by Ian Fleming, ISBN: 978-0470-74658-5.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Module 1	8
2.	Module 2	6
3.	Module 3	8
4.	Module 4	10
5	Module 5	8
Total number of hours		40